Maria Akter

Ph.D. Candidate in Mathematics

Department of Mathematics, The University of Alabama · Tuscaloosa, Alabama, USA

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Research Interests

My research lies in commutative algebra, primarily in positive characteristic. I am particularly interested in studying singularities through Frobenius methods, with an emphasis on F-signature, Frobenius splitting ratios, and the stability of F-singularities under perturbations. My broader research interests include differential operators, Frobenius compatible filtrations, and their applications to the study of singularities.

Education

The University of Alabama — Jan. 2022–present
Ph.D. in Mathematics (expected May 2027); Advisor: Thomas Polstra — Tuscaloosa, USA

Bangladesh University of Engineering and Technology (BUET) — 2015–2019
M.Phil. in Mathematics — Dhaka, Bangladesh

University of Dhaka — 2011–2013
M.Sc. in Pure Mathematics — Dhaka, Bangladesh

University of Dhaka — 2005–2010
B.Sc. in Mathematics — Dhaka, Bangladesh

Research Publications and Preprints

1. **Maria Akter**. *On $\mathfrak{m}$-adic Continuity of F-Splitting Ratio.* Preprint, 2025. arXiv:2505.12174. [Submitted]
2. S. M. Atiqur Rahman Chowdhury, Ashish Barmon, Sharmin Alam, and **Maria Akter**. *Numerical Simulation that Provides Non-oscillatory Solutions for Porous Medium Equation and Error Approximation of Boussinesq's Equation.* Universal Journal of Computational Mathematics **5**(2), 25–33 (2017). doi:10.13189/ujcmj.2017.050203.
3. **Maria Akter** and Salma Nasrin. *Invariant Differential Operator on Homogeneous Space.* International Journal of Mathematics and Computational Science **3**(1), 1–5 (2017).
4. **Maria Akter** and Salma Nasrin. *Symmetric Pair from Symmetric Space and Lie Group.* Journal of Engineering Mathematics and Statistics (2018).

Research in Progress

- **$\mathfrak{m}$-adic Stability of F-Singularities for F-Graded Systems**. Joint work with Anna Brosowsky [Manuscript in Preparation].
- **Monotonicity of F-Signature under Perturbations**. Joint work with Thomas Polstra.
- **Differential Operators and Frobenius Indecomposability**. Joint work with Thomas Polstra and Hongyu Luo.

- **Frobenius Maps, Filtrations, and Birational Modules**. Joint work with Thomas Polstra.

Academic Appointments

Graduate Research Assistant — May 2026–present
The University of Alabama — Tuscaloosa, USA

Graduate Teaching Assistant — Jan. 2022–Apr. 2026
The University of Alabama — Tuscaloosa, USA

Assistant Professor — June 2018–present
Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh

Lecturer — Feb. 2015–May 2018
Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh

Lecturer — May 2016–May 2017
Sonargaon University — Dhaka, Bangladesh

Lecturer — May 2014–Feb. 2015
BRAC University — Dhaka, Bangladesh

Lecturer — Aug. 2013–Dec. 2013
United International University — Dhaka, Bangladesh

Selected Research Talks and Presentations

Joint Mathematics Meetings, Special Session on Homological Methods, Syzygies, and Resolutions in Commutative Algebra, Chicago (scheduled) — Jan. 2027

Algebra Seminar, Syracuse University — Oct. 2026

URICA Conference, University of Nebraska, Lincoln — Aug. 2026

AMS Sectional Meeting, North Dakota State University, Fargo — Apr. 2026

Commutative Algebra Seminar, University of Utah — Nov. 2025

Commutative Algebra Reading Seminar (CARS), University of Nebraska–Lincoln — Oct. 2025

Commutative Algebra Seminar, University of Nebraska–Lincoln — Oct. 2025

Algebra Seminar, Auburn University — Sept. 2025

Algebra/Topology Seminar, The University of Alabama — Sept. 2025

Summer Camp, University of Nebraska–Lincoln (short talk) — Aug. 2025

CATS Conference, Georgia State University — Apr. 2025

WICA Conference, Fields Institute, Toronto (short talk) — Mar. 2025

GEMS Conference, University of Minnesota (short talk) — Nov. 2023

Poster Presentations

On $\mathfrak{m}$-adic Continuity of F-Splitting Ratio — Oct. 2025
KUMUNU, University of Missouri, Columbia, USA.

On $\mathfrak{m}$-adic Continuity of F-Splitting Ratio — Apr. 2025
CATS Conference, Georgia State University, USA.

On $\mathfrak{m}$-adic Continuity of F-Splitting Ratio — Mar. 2025
WICA Conference, Fields Institute, Toronto, Canada.

On Inversion of Adjunction of Splitting Dimensions — Feb. 2024
SLAM Conference, University of Oklahoma, USA.

Honors, Awards, and Fellowships

Hasell Palmer Scholarship for Fall 2026 and Spring 2027 — 2026
Department of Mathematics, The University of Alabama, Tuscaloosa, USA.

Henry Miller Fellowship for Fall 2026 and Spring 2027 — 2026
Department of Mathematics, The University of Alabama, Tuscaloosa, USA.

Outstanding Research by a Ph.D. Student Award — 2026
Department of Mathematics, The University of Alabama, Tuscaloosa, USA.

Nominated by The University of Alabama to attend the SLMath Summer Program — 2025
Fields Institute, Toronto, Canada.

Hasell Palmer Scholarship for Fall 2025 and Spring 2026 — 2025
Department of Mathematics, The University of Alabama, Tuscaloosa, USA.

Lucretia Anne Dehaney Endowed Scholarship — 2024
Department of Mathematics, The University of Alabama, Tuscaloosa, USA.

Highest GPA Award in the Third Year of Undergraduate Study — 2010
University of Dhaka, Dhaka, Bangladesh.

Teaching Experience

The University of Alabama

- MATH 238 — Applied Differential Equations I, Fall 2025
- MATH 110 — Finite Mathematics, Summer 2025
- MATH 237 — Linear Algebra, Spring 2025
- MATH 572 — Linear Algebra (grader and assistant instructor), Fall 2024
- MATH 238 — Applied Differential Equations I, Spring 2024
- MATH 126 — Calculus II, Fall 2023
- MATH 115 — Precalculus and Trigonometry, Fall 2022
- MATH 125 — Calculus I, Spring 2022

Bangladesh University of Engineering and Technology (BUET)

January 2015

- MATH 111 — Mathematics I
- MATH 141 — Calculus I
- MATH 243 — Advanced Calculus
- MATH 235 — Vector Analysis and Statistics
- MATH 263 — Complex Variables, Harmonic Analysis and PDE
- MATH 133 — Matrices and Coordinate Geometry

January 2016

- MATH 113 — Calculus
- MATH 161 — Differential Calculus, Solid Geometry and Vectors
- MATH 141 — Calculus I
- MATH 121 — Calculus
- MATH 191 — Differential and Integral Calculus
- MATH 137 — Differential and Integral Calculus, Matrices

January 2017

- MATH 113 — Calculus
- MATH 145 — Linear Algebra
- MATH 231 — Differential Equations
- MATH 237 — Laplace Transform and Vector Analysis
- MATH 121 — Calculus

January 2019

- MATH 213 — Differential Equations
- MATH 161 — Differential Calculus, Solid Geometry and Vectors
- MATH 271 — PDE and Vector Analysis
- MATH 171 — Calculus and Ordinary Differential Equations

July 2019

- MATH 191 — Differential and Integral Calculus
- MATH 161 — Differential Calculus, Solid Geometry and Vectors
- MATH 283 — Statistics, PDE and Matrices
- MATH 257 — Ordinary and Partial Differential Equations

BRAC University

Summer 2014

- MATH 123 — Calculus I
- MATH 121 — Basic Algebra
- MATH 211 — Calculus II

Fall 2014

- MATH 123 — Calculus I
- MATH 121 — Basic Algebra

- MATH 211 — Calculus II

United International University

Summer 2013

- MATH 134 — Calculus I
- MATH 121 — Basic Algebra

Sonargaon University

Fall 2016

- MATH 167 — Integral Calculus I

Spring 2017

- MATH 167 — Integral Calculus I

Fall 2017

- MATH 144 — Matrices

Academic Service and Leadership

Co-organizer, AMS Special Session on Recent Developments in Commutative Algebra — 2027 (scheduled)
University of Alabama at Birmingham; with Thomas Polstra and Anna Brosowsky.

Co-organizer, Dhaka Undergraduate Mathematics Olympiad Final Event — 2019
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh.

Co-organizer, Dhaka South Undergraduate Mathematics Olympiad — 2018
BUET, Dhaka, Bangladesh.

Co-organizer, Dhaka South Undergraduate Mathematics Olympiad — 2017
BUET, Dhaka, Bangladesh.

Secretary, BUGS — 2021
BUET, Dhaka, Bangladesh.

Examiner, Recruitment Examination (Water Recruit), Bangladesh — 2019
Dhaka, Bangladesh.

Member, BUET Entrance Examination Examining Committee — 2019
BUET, Dhaka, Bangladesh.

Member, BUET Entrance Examination Scrutiny Committee — 2017, 2018
BUET, Dhaka, Bangladesh.

Member, Graduate Teaching Assistant Recruitment Interview Board — 2018
BUET, Dhaka, Bangladesh.

Member, BRTC Recruitment Board — 2018–2021
BUET, Dhaka, Bangladesh.

Secretary, Mathematics Department Study Tour Organizing Committee — 2019
BUET, Dhaka, Bangladesh.

Selected Workshops and Professional Development

Summer Camp 2025 — Aug. 2025
University of Nebraska–Lincoln, Nebraska, USA.

An Introduction to Recent Trends in Commutative Algebra Workshop — June 2025
Fields Institute, Toronto, Canada.

SLMath 1060: Recent Developments in Commutative Algebra Workshop — Apr. 2024
SLMath, Berkeley, California, USA.

CATS Conference — Apr. 2023
The University of Alabama, Tuscaloosa, USA.

20th International Mathematics Conference — Dec. 2017
University of Dhaka, Dhaka, Bangladesh.

Training Workshop on Teaching and Learning for IQAC — Nov. 2016
BUET, Dhaka, Bangladesh.

Faculty Orientation Workshop on Teaching and Learning — Feb. 2014
BRAC University, Dhaka, Bangladesh.

Professional Memberships

- Member, Association for Women in Mathematics — 2022–present
- Member, Association of Bangladeshi Women in Mathematics — 2021–present
- Life Member, BUET Mathematics Alumni Association — Since 2020
- Life Member, DUMAA — Since 2013
- Life Member, Bangladesh Mathematical Society — Since 2013